

MedGraphRAG: Integrated Tri-Modal Retrieval Architecture and Sentence-Level Claim Grounding for Clinical Oncology Question Answering

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Abstract

Large Language Models (LLMs) exhibit strong general-purpose text generation capabilities, but their unconstrained deployment in clinical decision-support contexts is constrained by hallucination, lack of provenance, and unreliable grounding in verified medical literature. Rather than proposing novel individual algorithms, this paper presents **MedGraphRAG**, a specific integrated architecture and empirical evaluation framework designed for evidence-grounded oncology question answering. MedGraphRAG unifies three established retrieval channels: high-dimensional dense vector search using **BAAI/bge-base-en-v1.5** in ChromaDB, sparse lexical keyword search via BM25Okapi with SciSpaCy query entity expansion, and multi-hop Knowledge Graph traversal structured in NetworkX with Inverse Entity Frequency (IEF) topological distance decay scoring. Candidate passages undergo Min-Max score standardization, Jaccard token deduplication ($\tau = 0.65$), and Cross-Encoder reranking (**BAAI/bge-reranker-base**). A local, 4-bit quantized **llama3.2:latest** (3.8B) generator synthesizes clinical answers under a 10-rule constrained prompt, combined with a sentence-level Natural Language Inference (NLI) entailment claim checker ($\tau = 0.70$) and automatic citation generation. On a benchmark of 200 curated oncology question–answer pairs drawn from six oncology textbooks (944 indexed chapters, 15,935 retrieval chunks), MedGraphRAG is evaluated across five 100-question ablation conditions ($N=500$ evaluations). The full system achieves a Precision@5 of 0.8950, a Faithfulness score of 0.9080, a Hallucination rate of 0.0920 (versus 0.3913 for the dense baseline), and statistically significant improvements over the baseline ($p_{\text{adj}} = 3.89 \times 10^{-17}$ for Precision@5). A 30-item clinician alignment study involving three clinical oncology specialists demonstrates strong Human–LLM evaluation alignment ($r = 0.9315$ for Faithfulness, $r = 0.9556$ for Groundedness).

Index Terms: Retrieval-Augmented Generation, Integrated Systems, Medical Question Answering, Clinical Natural Language Processing, CrossEncoder Reranking, Large Language Models, Llama-3.2, Evidence Grounding, Sentence Embeddings, Clinician Alignment.

I. Introduction

Large Language Models (LLMs) have demonstrated increasingly strong performance on open-domain and biomedical text generation tasks [1]–[3], and specialized medical LLMs have shown that clinical knowledge can be encoded and elicited from sufficiently large models [4]. However, the unconstrained deployment of LLMs in clinical decision-support contexts remains limited by three interrelated problems. First, *medical hallucination* – the generation of fluent but factually unsupported or incorrect clinical statements – poses a direct risk in any setting where the output could plausibly influence patient care [5]. Second, purely parametric generation offers no provenance: a clinician has no mechanism to trace a generated claim back to a specific, verifiable source in the medical literature. Third, standard LLM inference lacks *evidence grounding*, i.e., an explicit constraint that every generated statement be derivable from a retrieved, citable passage rather than from the model's internal parameters alone.

Retrieval-Augmented Generation (RAG) [6] addresses these problems in part by conditioning generation on passages retrieved from an external corpus at inference time, reducing (though not eliminating) hallucination and enabling citation. However, conventional RAG pipelines typically perform a single-pass, flat dense retrieval step over an undifferentiated chunk index [7], [8].

To address these challenges, we present **MedGraphRAG**. Rather than claiming the invention of individual components, our contribution lies in the **design, integration, and systematic empirical evaluation** of a tri-modal hybrid retrieval engine and post-generation sentence-level claim verification framework tailored specifically for medical oncology QA.

Primary Contributions

- 1. Integrated Tri-Modal Retrieval Architecture:** A unified system combining dense vector search (**BAAI/bge-base-en-v1.5**), sparse BM25 lexical search with SciSpaCy query expansion, and NetworkX Knowledge Graph traversal with topological distance decay.
- 2. Evidence-Grounded Generation & NLI Verification Stage:** A joint generation and verification framework combining a 10-rule constrained system prompt ($\tau_{\text{refusal}} = 0.35$) with a sentence-level NLI entailment claim verifier ($\tau_{\text{grounding}} = 0.70$) that attaches structured provenance cards (98.5% coverage).
- 3. Empirical Benchmark & Statistical Significance:** A full 5-way component ablation study over 100 stratified gold questions ($N=500$ evaluations), applying paired two-sided Wilcoxon signed-rank testing with Holm-Bonferroni correction ($p_{\text{adj}} = 3.89 \times 10^{-17}$ for Precision@5).
- 4. Clinician & Dual-Judge Alignment Study:** A 30-item clinical expert alignment evaluation involving three board-certified oncology specialists, demonstrating strong Human–LLM evaluation correlation ($r = 0.9315$ for Faithfulness, $r = 0.9556$ for Groundedness).

II. Related Work & System Comparison

Table I: Comparison of MedGraphRAG Against Representative Retrieval and RAG Architectures

System	Retrieval Channels	Domain	Reranking Engine	Grounding / Citation
BM25 [19]	Sparse, flat	General	None	None
DPR [7]	Dense, flat	Open domain	None	None
RAG [6]	Dense, flat	Open domain	None	Implicit
GraphRAG [18]	Graph-based	General	None	Partial
Clinfo.ai [17]	Dense, flat	Clinical lit.	None	Yes (inline)
MedGraphRAG (Proposed)	Dense + Sparse + Graph	Oncology	BAAI CrossEncoder	Structured Cards w/ Confidence

III. Proposed Method

A. Tri-Modal Retrieval Formulations

Dense vector retrieval utilizes **BAAI/bge-base-en-v1.5** ($E(q)$) with normalized cosine similarity:

$E(q)$	$\cdot$	$E(c)$	$\} = E(q) \cdot E(c), \text{ } \lrcorner$	$E(\cdot)$
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Multi-hop Knowledge Graph traversal extracts query entities $\mathcal{E}_q$ via SciSpaCy (**en_core_sci_sm**) and expands up to $H=2$ hops in NetworkX, scoring chunks via Inverse Entity Frequency (IEF) shortest-path topological decay:

$$S_{\text{graph}}(q, c) = \max_{e \in \mathcal{E}_q} \left(\sum_{v \in \mathcal{N}_H(e)} \text{ChunkEntities}(c) \cdot \frac{1}{1.0 + \text{dist}_G(e, v)} \right)$$

Scores are Min-Max standardized and fused:

$$S_{\text{hybrid}}(c) = 0.35 \cdot S_{\text{norm_dense}}(c) + 0.30 \cdot S_{\text{norm_bm25}}(c) + 0.35 \cdot S_{\text{norm_graph}}(c)$$

B. CrossEncoder Reranking & Sentence-Level NLI Verification

Candidate passages are reranked using **BAAI/bge-reranker-base**:

$$S_{\text{rerank}}(q, c) = \text{CrossEncoder}(q, c)$$

Given answer $A = \{s_1, \dots, s_M\}$, sentence entailment against context P is evaluated:

$$\text{LexicalOverlap}(s_j, P) = 0.4 \cdot \text{LexicalOverlap}(s_j, P) + 0.6 \cdot \cos(E(s_j), E(P)) \geq 0.70$$

IV. Results & Clinician Alignment Study

Table II: Empirical Benchmark & Component Ablation Results (\$N=100\$ Questions, \$N=500\$ Evaluations)

Method	Ret.Acc.	P@5	R@5	Faithf.	Ans.Rel.	Ground.	Halluc.↓	Lat.(s)
Exp. 1 (Dense Only)	0.8000	0.4100	0.9507	0.6087	0.7341	0.6717	0.3913	14.22
Exp. 2 (No Graph)	0.9200	0.4640	0.9700	0.6964	0.8426	0.7550	0.3036	25.40
Exp. 3 (No BM25)	0.8600	0.4080	0.9596	0.6738	0.8411	0.6383	0.3262	31.47
Exp. 4 (No Reranker)	0.8500	0.3280	0.9716	0.6838	0.8358	0.6842	0.3162	18.14
Exp. 5 (Full MedGraphRAG)	0.9300	0.8950	0.9776	0.9080	0.9150	0.9120	0.0920	25.57

Table III: Clinician Expert & Dual-Judge Agreement Statistics (\$N=30\$ Clinical Subsample)

Evaluation Dimension	Inter-Judge (Primary vs Strong)	Human–LLM Alignment
Faithfulness Pearson \$r\$	0.8282	0.9315
Faithfulness Cohen's \$\kappa\$	1.0000	1.0000
Groundedness Pearson \$r\$	0.9339	0.9556
Groundedness Cohen's \$\kappa\$	1.0000	1.0000

V. Discussion & Conclusion

Causal Mechanism Analysis

The empirical results demonstrate that Precision@5 gains are primarily driven by Cross-Encoder reranking (0.3280 → 0.8950), while Groundedness is enhanced by Knowledge Graph expansion (0.7550 → 0.9120). The reduction in Hallucination Rate (0.3913 → 0.0920) is a joint outcome of the evidence-grounded prompting and sentence NLI claim verification stage.

Conclusion

MedGraphRAG presents an integrated tri-modal retrieval architecture and sentence-level claim verification framework for clinical oncology QA. The system achieves high precision (0.8950), faithfulness (0.9080), and strong clinician evaluation alignment (\$r = 0.9315 - 0.9556\$), providing a defensible foundation for evidence-grounded medical AI systems.